RETAIL SALES PROJECT REPORT

1. Introduction

This project involved the analysis of a retail sales dataset using Microsoft Excel. The aim was to demonstrate how Excel can be used not only for basic calculations, but also for data validation, automation, data modelling, analysis, visualization, and reporting.

Several advanced Excel features were applied throughout the project, including Data Validation, Excel functions, Macros, Power Pivot, Pivot Tables, Pivot Charts, Dashboard visualization, Box & Whisker analysis, and the Lookup function.

The completed analysis provides an overview of sales performance and customer purchasing patterns across product categories, gender, age groups, and months.

2. Objectives of the Project

The objectives of the project were to:

- Validate and control data entry using Excel's Data Validation feature.
- Apply conditional Excel functions to analyze sales data.
- Use Macros to automate a repetitive analysis task.
- Use Power Pivot to create a relational data model.
- Create relationships between different tables.
- Use Pivot Tables to summarize the dataset.
- Create Pivot Charts to visualize the results.
- Develop an interactive Dashboard containing multiple visuals.
- Use a Box & Whisker plot to analyze the distribution of sales.
- Apply the LOOKUP function to categorize customers according to age.
- Interpret the results and provide useful business insights.

3. Dataset Description

The dataset contains 1,000 retail transactions.
The main columns are:

- Transaction ID
- Date
- Customer ID
- Gende
- Age
- Age Group
- Product Category
- Quantity
- Price per Unit
- Sales

The three product categories are:

- Beauty
- Clothing
- Electronics

The dataset contains 2,514 units sold and total sales of $456,000.

4. Data Preparation
Before carrying out the analysis, the dataset was reviewed and prepared for use.
The data was organized into appropriate worksheets and different Excel features were applied to improve its reliability and usability.
Data Validation was used to restrict incorrect entries, while formulas and lookup functions were used to support the analysis.
The dataset was also structured into separate tables for the Power Pivot model.

5. Data Validation
Data Validation was used to control the type and format of information that could be entered into important fields in the data.

- **Transaction ID**

The Transaction ID was required to follow the format:
One letter + four numbers
Example:
T0001
An invalid entry such as TT001 was tested and rejected by Excel.
This demonstrated that the validation rule was functioning.

- **Customer ID**

Customer IDs were required to follow the format:
CUS0001
An invalid entry such as CU001 was tested and rejected.

- **Date Validation:**

The Date column was restricted to:
Dates that are not in the future and weekdays only.
A future date was tested and rejected.
A Saturday date was also tested and rejected.
This confirmed that both date restrictions were working.

- **Product Category Validation**

The Product Category field was restricted to text entries.
A numerical value such as 12345 was entered as a test and Excel rejected it.
This helped maintain consistency in the Product Category column.

The Data Validation exercise helped improve the accuracy, consistency, and reliability of data entry.

6. Excel Functions
Six conditional functions were applied to the dataset:
These functions were used to answer different questions about sales and customer's behavior.

- SUMIF was used to calculate sales based on a single condition.

- SUMIFS was used when more than one condition was required.

- COUNTIF was used to count records meeting one condition.
- COUNTIFS was used where multiple conditions were required.

- AVERAGEIF was used to calculate average sales based on one condition

- AVERAGEIFS was used to calculate averages using multiple conditions.

The functions demonstrated the ability to perform both simple and multi-condition analysis within Excel.

7. Macros
A Macro was created to automate the analysis of sales by product category.
The Macro generated a Sales by Category summary and corresponding chart.

The use of a Macro demonstrates how repetitive Excel tasks can be automated, saving time and reducing manual work.

The Macro was particularly useful because instead of manually creating the summary and chart each time, the task could be automated.

8. Power Pivot and Data Modelling
Power Pivot was used to create a relational data model for the retail sales data.
The model consisted of four main tables:

- **FactSales**

This table contains the transaction-level sales information which are:
Transaction ID
Date
Customer ID
Product Category
Quantity
Price per Unit
Sales

- **DimCustomer**

This table contains customer-related information, including:
Customer ID
Gender
Age

- **DimProduct**

This table contains unique product categories:
Beauty
Clothing
Electronics
Duplicate product categories were removed so that the table could function correctly as a dimension table.

- **DimDate**

The Date dimension contains:

Date
Year
Month
Month Number

Relationships
Relationships were created between the dimension tables and FactSales.
The relationships were:

- DimCustomer – FactSales: Customer ID was used to connect the tables.
- DimDate – FactSales: Date was used to connect the tables.
- DimProduct – FactSales: Product Category was used to connect the tables.

The relationships were established as Many-to-One relationships, with the dimension tables on the one side and FactSales on the many sides.

Total Sales Measure
A Power Pivot measure was created:
This measure calculates total sales within the Power Pivot data model.
The resulting total sales were:
456,000

Power Pivot made it possible to organize the data into related tables instead of keeping all information in one large table. This creates a more structured model and makes the dataset easier to analyse.

9. Pivot tables Charts and Dashboard
Pivot Charts were created from the Pivot Tables to make the analysis easier to understand visually.
The Dashboard contained at least six major visuals, which including:

- Total Sales, Total Quantity and Number of Transactions
- Sales by Product Category
- Monthly Sales Trend
- Sales by Gender
- Sales Share by Category
- Total Quantity Sold by Category
- Sales by Age Group
- Interactive slicers and a timeline filter.

The dashboard provides a summarized view of the retail business performance and allows users to explore the information visually.

10. Box & Whisker Plot
A Box & Whisker plot was created to examine the distribution of sales.
The median of 135 represents the middle of the ordered sales observations. The large difference between Q1 and Q3 indicates substantial variation among the middle 50% of sales transactions. The distribution is also strongly influenced by higher sales values.

The Box & Whisker analysis showed considerable variation in transaction sales, with values ranging from 25 to 2,000.

11. LOOKUP Function

The LOOKUP function was used to categorize customers according to their age.
A lookup table was created containing the starting age for each age group.
The lookup function allowed individual ages to be converted into meaningful age-group categories, making the data easier to analyze in Pivot Tables and charts.

12. Key Findings

The Excel analysis produced several important findings.

- **Overall Sales**

Total sales across the 1,000 transactions were:
456,000
with 2,514 units sold.

- **Product Performance**

Electronics generated the highest sales of 156,905 followed by Clothing with 155,580, while Beauty recorded 143,515.

- **Gender**

Female customers generated 232,840 in sales, compared with 223,160 generated by male customers.

- **Monthly Performance**

May was the strongest month with sales of 53,150 while September was the weakest month with sales of 23,620.

- **Quantity**

Clothing had the highest number of units sold at 894 units.

13. Recommendations

Based on the analysis, the following recommendations can be made:

- Focus on high-performing categories: Electronics generated the highest sales, so the business could continue monitoring this category and ensure adequate stock availability.
- Investigate low-performing periods: September had the lowest monthly sales. The business could investigate possible reasons for the decline and consider targeted promotions during weaker periods.
- Monitor quantity and revenue together: Clothing had the highest quantity sold, but Electronics generated the highest revenue. This shows why both sales value and quantity should be monitored.
- Use customer segmentation: Age-group and gender analysis can help the business understand its customer base and develop more targeted marketing strategies.
- Continue using dashboards: The dashboard provides a convenient way for decision-makers to monitor sales performance and identify trends.

14. Conclusion

This project demonstrated how Microsoft Excel can be used as a powerful tool for retail sales analysis.

The project covered several important Excel capabilities, including Data Validation, conditional functions, Macros, Power Pivot, data modelling, Pivot Tables, Pivot Charts, Dashboard creation, Box & Whisker analysis, and LOOKUP functions.

The analysis showed that the dataset generated total sales of 456,000 from 1,000 transactions, with Electronics recording the highest category sales and May recording the highest monthly sales.

The project also demonstrated the importance of data quality, automation, structured data modelling, and visual presentation when analyzing business data.

Overall, the project provided practical experience in using Excel to transform raw retail data into meaningful information that can support business decision-making.